Problem H. Matriz

Time Limit 2000 ms

Problem Statement

You are given a positive integer N.

Fill each square of a grid with N rows and N columns by writing a positive integer not greater than N^2 so that all of the following conditions are satisfied.

- Two positive integers written in horizontally or vertically adjacent squares never sum to a prime number.
- Every positive integer not greater than N^2 is written in one of the squares.

Under the Constraints of this problem, it can be proved that such a way to fill the grid always exists.

Constraints

- $3 \leq N \leq 1000$

Input

The input is given from Standard Input in the following format:

```
N
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Output

Print a way to fill the grid under the conditions in the following format, where A_{ij} is the positive integer at the i-th row and j-th column:

$$
\begin{array}{l}
A_{11} \ \ldots \ A_{1N} \\
\vdots \\
A_{N1} \ \ldots \ A_{NN}
\end{array}
$$

If there are multiple ways to fill the grid under the conditions, any of them will be accepted.

Sample 1

Input	Output
4	15 11 16 12 13 3 6 9 14 7 8 1 4 2 10 5

In this grid, every positive integer from 1 through 16 is written once. Additionally, among the sums of two positive integers written in horizontally or vertically adjacent squares are $15 + 11 = 26$, $11 + 16 = 27$, and $15 + 13 = 28$, none of which is a prime number.